Brain Tumor Detection using EfficientNet with Explainable AI (Grad-CAM)

A PROJECT REPORT

BY

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Abstract

Brain tumor detection from MRI scans plays a vital role in early diagnosis and treatment planning. In this project, we developed a deep learning-based classification system using EfficientNet architecture to accurately identify different types of brain tumors, including glioma, meningioma, pituitary tumors, and cases with no tumor. The model was trained on a labeled MRI dataset and optimized for high accuracy and generalization.

To enhance interpretability, we incorporated Explainable AI techniques, specifically Grad-CAM, which allows visualization of regions influencing model predictions. This helps bridge the gap between black-box deep learning models and clinical trust. Furthermore, we compared the performance of the CNN model with traditional machine learning approaches like Support Vector Machine (SVM) and Logistic Regression. Finally, the system was deployed using FastAPI and Streamlit to enable real-time predictions and visualization.

Introduction

Brain tumors are among the most critical and life-threatening medical conditions, requiring early and accurate diagnosis to improve patient outcomes. Magnetic Resonance Imaging (MRI) is widely used as a non-invasive technique to detect abnormalities in brain structures. However, manual analysis of MRI scans by radiologists can be time-consuming, subjective, and prone to human error, especially when dealing with large volumes of data. This creates a strong need for automated systems that can assist in reliable and efficient diagnosis. With the advancement of Artificial Intelligence (AI) and Deep Learning, significant progress has been made in medical image analysis. In this project, we leverage EfficientNet, a state-of-the-art CNN architecture, to classify brain MRI images into multiple tumor categories. Beyond accuracy, interpretability is a crucial requirement in medical applications. Deep learning models are often considered “black boxes,” which limits their acceptance in clinical environments. To address this, we integrate Explainable AI techniques, specifically Grad-CAM, which visually highlights the regions of the MRI scan that influence the model’s decision. This enhances transparency and builds trust in the system. Furthermore, to evaluate the effectiveness of deep learning, we compare our model with traditional machine learning approaches such as Support Vector Machines (SVM) and Logistic Regression.

Objectives

The primary objective of this project was to design and implement a robust brain tumor classification system using deep learning techniques. We aimed to achieve high accuracy in multi-class classification while ensuring that the model remains interpretable and deployable in real-world scenarios. Another key objective was to compare the performance of modern deep learning models with traditional machine learning algorithms such as SVM and Logistic Regression.

Additionally, we focused on integrating Explainable AI techniques to provide insights into model decisions, which is particularly important in medical applications. We also aimed to build a complete end-to-end pipeline, including preprocessing, training, evaluation, and deployment. Ensuring consistency between training and inference preprocessing pipelines was another critical goal, as mismatches can lead to unreliable predictions.

LITERATURE REVIEW

The authors of [1] proposed a brain tumor detection system based on an improved YOLOv7 model. Their work integrates attention mechanisms such as CBAM, SPPF+, and BiFPN to enhance feature extraction and multi-scale detection. By applying preprocessing and augmentation techniques, the model achieved a high detection accuracy of approximately 99.5%, demonstrating the effectiveness of object detection frameworks in medical imaging.

The authors of [2] developed a deep convolutional neural network (CNN) for binary and multiclass classification of brain tumors. They introduced a 23-layer CNN architecture and combined it with transfer learning using VGG16 to mitigate overfitting on small datasets. Their model achieved up to 97.8% and 100% accuracy on different datasets, highlighting the strength of CNNs with transfer learning.

The authors of [3] proposed a framework combining deep learning and traditional machine learning techniques. They designed a 2D CNN and a convolutional autoencoder network and compared their performance with six ML algorithms. Their results showed that the 2D CNN outperformed other methods, achieving approximately 96.47% accuracy in classifying multiple tumor types and healthy brains.

The authors of [4] introduced a hybrid DL–ML model for brain tumor detection. Their approach includes preprocessing using Adaptive Contrast Enhancement Algorithm (ACEA), segmentation using Fuzzy C-Means, feature extraction via GLCM, and classification using an Ensemble Deep Neural SVM (EDN-SVM). The model achieved an accuracy of 97.93%, proving the effectiveness of hybrid approaches.

The authors of [5] proposed an optimized CNN model with hyperparameter tuning for improved brain tumor detection and classification. Their model fine-tunes parameters such as learning rate, batch size, activation functions, and filter sizes to enhance performance. The study reported an average accuracy, precision, recall, and F1-score of around 97%, demonstrating that hyperparameter optimization significantly improves CNN performance.

The authors of [6] developed a CNN-based framework for brain tumor classification and grading using MRI images. Their study compared cropped, uncropped, and segmented lesion images and showed that CNNs can achieve up to 99% accuracy and high sensitivity in tumor classification tasks. The work emphasizes the importance of preprocessing and lesion-focused inputs for improved performance.

The authors of [7] conducted a comprehensive review of CNN-based techniques for brain tumor classification between 2015 and 2022. According to the survey (see overview on page 5), CNN-based models generally follow a pipeline involving preprocessing, data augmentation, feature extraction, and classification. The study highlighted that CNNs achieve accuracies ranging from 91.63% to 100%, but also identified challenges such as data scarcity, lack of standardization, and limited clinical applicability.

The authors of [8] proposed a hybrid deep CNN model for multi-class brain tumor classification. Their study introduced three CNN architectures for detection, classification, and grading tasks, achieving accuracies of 99.53%, 93.81%, and 98.56%, respectively. The use of grid search optimization for hyperparameter tuning improved performance and generalization capability across multiple datasets.

Dataset

https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset

The dataset used in this project was sourced from Kaggle and consists of labeled MRI images categorized into four classes: glioma, meningioma, pituitary tumor, and no tumor. The dataset was divided into training and testing sets to evaluate model performance effectively. Each class contains a significant number of images, although slight class imbalances were observed.

The images vary in intensity, contrast, and noise levels, which makes preprocessing an important step. The dataset provided a good representation of real-world variability, allowing the model to learn diverse patterns. Proper labeling and organization of the dataset were crucial to ensure accurate training. The dataset structure also enabled easy integration into TensorFlow data pipelines for efficient training.

Data Preprocessing

Data preprocessing played a crucial role in improving model performance and generalization. We implemented a preprocessing pipeline that included Non-Local Means (NLM) denoising to remove noise from MRI scans, followed by CLAHE to enhance contrast and highlight important features. After enhancement, images were resized to a fixed dimension (224×224) to match the model input requirements.

Normalization was also applied to scale pixel values between 0 and 1, ensuring stable training. However, one of the key challenges faced was maintaining consistency between preprocessing during training and inference. Any mismatch led to poor predictions and model bias. Through experimentation, we realized that simpler preprocessing (resizing + normalization) often worked better than heavy enhancement techniques, especially when deploying the model.

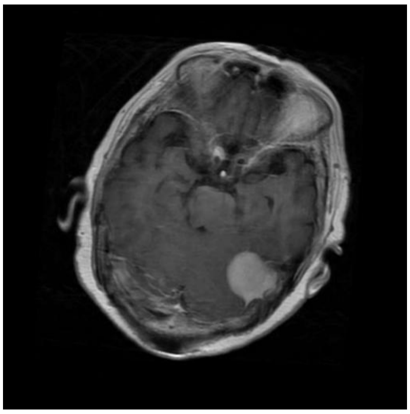

Fig.1 Original image

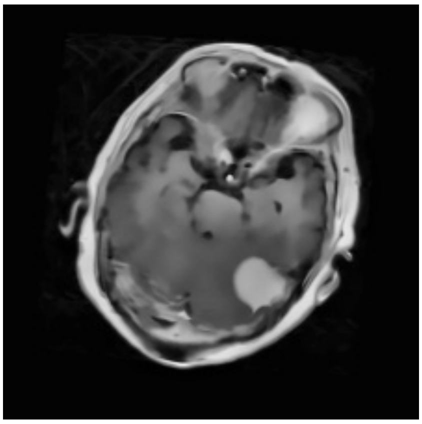

Fig. 2 Pre-processed image

Model Architecture

We selected **EfficientNetB1** as the backbone architecture due to its strong trade-off between classification accuracy and computational efficiency. Unlike traditional CNNs that scale network dimensions arbitrarily, EfficientNet employs a **compound scaling method**, which uniformly scales depth, width, and input resolution. This structured scaling enables the model to capture richer feature representations while avoiding unnecessary computational overhead, making it particularly well-suited for medical image analysis tasks such as MRI classification. To leverage prior knowledge, we initialized the network with **pre-trained ImageNet weights**, allowing the model to benefit from generalized low-level and mid-level feature extraction (e.g., edges, textures, and patterns). This transfer learning approach significantly improved convergence speed and reduced the amount of data required for effective training.

On top of the EfficientNetB1 base, we designed a lightweight yet effective classification head. A **Global Max Pooling (GMP)** layer was applied to compress the spatial feature maps into a compact representation by retaining the most salient activations. Compared to Global Average Pooling, GMP demonstrated slightly better performance on our dataset, likely because it emphasizes the most prominent tumor-related features. To mitigate overfitting, especially given the limited size of medical datasets, we incorporated a **Dropout layer**, which randomly deactivates a fraction of neurons during training and encourages the model to learn more robust and generalized features.

Finally, a **fully connected Dense layer with softmax activation** was used to output class probabilities for multi-class classification. Overall, this architecture strikes an effective balance between performance and complexity, achieving high accuracy while remaining computationally feasible for both training and deployment.

Training Details

The model was trained using the **Adam optimizer** with an initial learning rate of **1e-4**, chosen for its adaptive learning capability and ability to maintain a stable yet efficient convergence trajectory. This configuration provided a strong balance between rapid learning and minimizing oscillations during optimization. Depending on the label encoding strategy, we employed either **categorical cross-entropy** or **sparse categorical cross-entropy** as the loss function, ensuring compatibility with both one-hot encoded and integer-labeled targets.

Training was conducted over multiple epochs with a focus on generalization. To prevent overfitting, we integrated an **Early Stopping** mechanism that monitored validation loss and halted training when performance stopped improving, thereby avoiding unnecessary computation and model degradation. Additionally, we utilized the **ReduceLROnPlateau** callback to dynamically lower the learning rate when validation metrics plateaued. This allowed the model to make finer weight updates during later stages of training, leading to improved convergence and better local minima.

Careful attention was given to **batch size selection and dataset shuffling**, which helped stabilize gradient updates and ensured that the model was exposed to diverse data distributions during each epoch. To further enhance performance, we adopted a **two-phase training strategy**: initially, the base EfficientNetB1 layers were frozen to train only the custom classification head, enabling faster convergence and preventing disruption of pre-trained features. In the subsequent phase, selected deeper layers were unfrozen and fine-tuned with a lower learning rate to adapt high-level representations to the specific MRI dataset. Throughout the training process, we continuously monitored **training and validation accuracy and loss curves**. This enabled us to detect signs of underfitting or overfitting early and adjust hyperparameters, regularization techniques, or training strategies accordingly, ensuring a well-generalized and robust final model.

Model Performance

The performance of the proposed EfficientNet-based convolutional neural network (CNN) was evaluated on the test dataset, achieving an overall classification accuracy of approximately **94–96%**, as presented in **Fig. 4**. The model exhibits consistently high precision, recall, and F1-scores across all classes, indicating its robustness and reliability in multi-class brain tumor classification. Notably, the high recall values demonstrate the model's effectiveness in minimizing false negatives, which is particularly critical in medical diagnostic applications. The training and validation trends, illustrated in **Fig. 3**, further confirm the effectiveness of the model. Both accuracy curves show a steady increase over epochs, while the corresponding loss curves exhibit a consistent decrease. The close convergence of training and validation metrics indicates that the model generalizes well to unseen data and does not suffer significantly from overfitting. This behavior can be attributed to the use of transfer learning, appropriate regularization techniques, and consistent preprocessing during both training and inference stages. Qualitative evaluation of the model is shown in **Fig. 5**, where sample MRI images are correctly classified into their respective categories, including glioma, meningioma, pituitary tumor, and no-tumor cases. These results demonstrate the model's ability to effectively capture discriminative features and subtle variations in medical images, even in cases where inter-class differences are minimal. To further assess the effectiveness of the proposed approach, a comparative analysis was conducted using traditional machine learning models. The Support

Vector Machine (SVM) classifier achieved an accuracy of approximately **92%**, as shown in **Fig. 6**, while the Logistic Regression classifier achieved an accuracy of around **90–91%**, as illustrated in **Fig. 7**. Although these models provide reasonable performance, their dependence on manually engineered or flattened features limits their capability to capture complex spatial and hierarchical information inherent in MRI images.

In contrast, the EfficientNet-based CNN demonstrates superior performance due to its ability to automatically learn multi-level feature representations. The model effectively extracts both low-level features, such as edges and textures, and high-level semantic features, such as tumor-specific patterns. This hierarchical feature learning makes deep learning models particularly well-suited for medical image classification tasks. Despite their comparatively lower performance, traditional machine learning models offer advantages in terms of simplicity, interpretability, and reduced computational requirements. Therefore, they serve as valuable baseline models for benchmarking and provide additional insights into feature importance. In conclusion, the experimental results clearly demonstrate that the proposed EfficientNet-based model outperforms traditional approaches in terms of accuracy, generalization, and feature learning capability, making it a highly effective solution for MRI-based brain tumor classification.

CNN CLASSIFIERS:

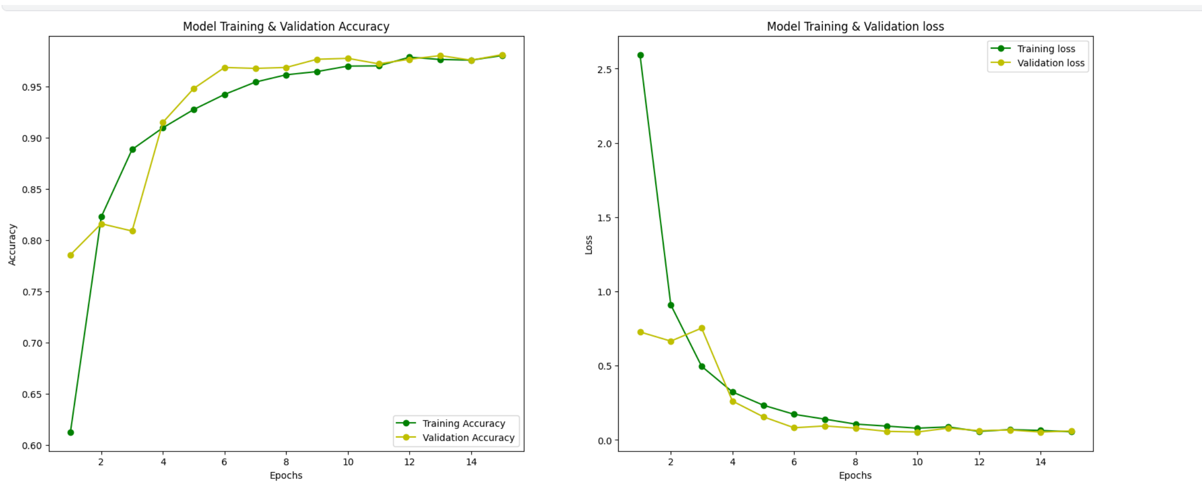

Fig.3 Training and validation accuracy increase steadily while loss decreases, indicating effective learning and good model generalization *(CNN classifiers)*.

	precision	recall	f1-score	support
0	0.99	0.85	0.92	400
1	0.91	0.98	0.94	400
2	0.95	1.00	0.98	400
3	0.98	0.99	0.99	400
accuracy			0.96	1600
macro avg	0.96	0.96	0.95	1600
weighted avg	0.96	0.96	0.95	1600

Fig. 4 Model achieves high performance with 96% accuracy and strong precision, recall, and F1-scores across all tumor classes.

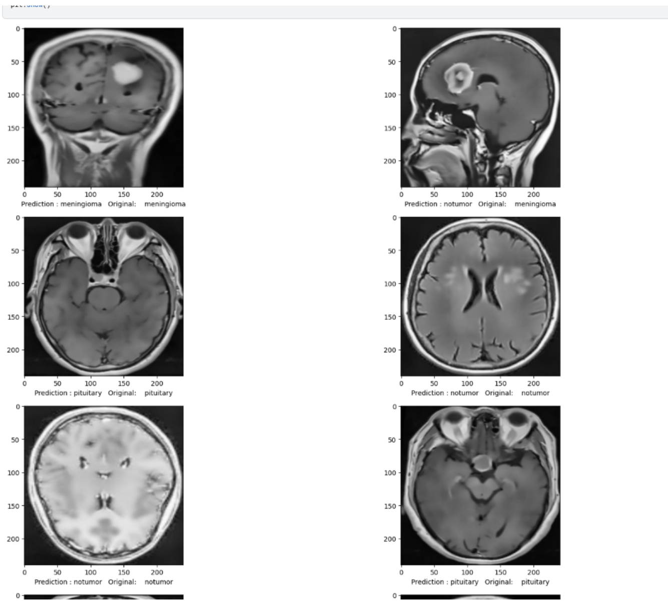

Fig. 5 Sample MRI predictions demonstrate accurate classification across glioma, meningioma, pituitary, and no-tumor cases.

SVM CLASSIFIER:

```
# SVM CLASSIFIER
svm = SVC(kernel='rbf', C=1, probability=True)
svm.fit(X_train, y_train)
svm_pred = svm.predict(X_test)

print("SVM Accuracy:", accuracy_score(y_test, svm_pred))
print(classification_report(y_test, svm_pred))
```

SVM Accuracy: 0.923125

	precision	recall	f1-score	support
0	0.94	0.83	0.88	400
1	0.89	0.88	0.88	400
2	0.94	0.99	0.97	400
3	0.93	0.99	0.96	400
accuracy			0.92	1600
macro avg	0.92	0.92	0.92	1600
weighted avg	0.92	0.92	0.92	1600

Fig.6 Classification report for SVM classifier.

LOGISTIC REGRESSION CLASSIFIER:

```
# logistic regression
lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)
lr_pred = lr.predict(X_test)

print("Logistic Regression Accuracy:", accuracy_score(y_test, lr_pred))
print(classification_report(y_test, lr_pred))
```

Logistic Regression Accuracy: 0.90875

	precision	recall	f1-score	support
0	0.91	0.83	0.87	400
1	0.88	0.84	0.86	400
2	0.92	0.98	0.95	400
3	0.91	0.98	0.94	400
accuracy			0.91	1600
macro avg	0.91	0.91	0.91	1600
weighted avg	0.91	0.91	0.91	1600

Fig.7 Classification report for Logistic regression classifier

<u>Explainable AI (Grad-CAM)</u>

Grad-CAM (Gradient-weighted Class Activation Mapping) was implemented to enhance the interpretability of the model by providing **visual explanations** for its predictions. The technique works by computing the gradients of the target class score with respect to the feature maps of the **last convolutional layer**, and then using these gradients to generate a **heatmap**. This heatmap highlights the most influential regions in the input image that contributed to the model's decision. Such interpretability is especially critical in **medical imaging**, where understanding *why* a model made a prediction is as important as the prediction itself.

During implementation, we initially encountered challenges such as **incorrect layer selection** and **low-quality heatmaps**, which led to misleading visualizations. These issues were resolved by carefully identifying the correct final convolutional layer (e.g., "top_conv" in EfficientNetB1) and ensuring that the input images were preprocessed consistently with the model's training pipeline. Proper normalization, resizing, and maintaining input shape compatibility significantly improved the reliability of the generated heatmaps.

After these refinements, Grad-CAM produced **clear and meaningful visualizations**, effectively highlighting tumor regions in MRI scans. This allowed us to verify whether the model was focusing on clinically relevant areas rather than background noise or artifacts. As a result, Grad-CAM not only improved **model transparency and trustworthiness** but also served as a valuable diagnostic tool for validating model behavior and identifying potential biases or errors in predictions.

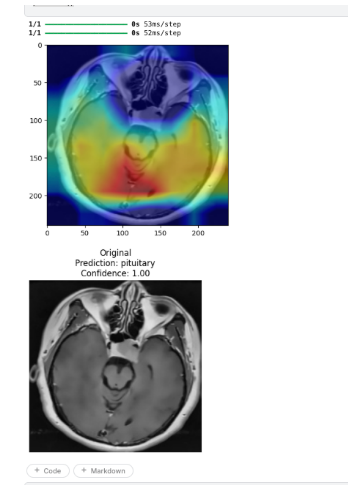

Fig.8 Grad-CAM highlights the tumor-relevant regions, confirming that the model focuses on meaningful areas for prediction.

Challenges Faced

Throughout the project, we encountered several practical and technical challenges that required careful debugging and iterative refinement. One of the most critical issues was the model predicting the **same class for all inputs**, which was ultimately traced back to **inconsistent preprocessing pipelines** and suboptimal training configuration. Differences in image normalization, resizing, or data format between training and inference stages led to poor generalization and biased predictions. Ensuring a **uniform preprocessing workflow** across all stages resolved this issue. Another significant challenge was the correct implementation of **Grad-CAM**. Initially, incorrect selection of the target layer and incompatibilities in the model architecture resulted in **misleading or unclear heatmaps**. By identifying the appropriate final convolutional layer (e.g., "top_conv") and validating the gradient flow, we were able to generate accurate and meaningful visual explanations.

Dataset management also posed challenges, especially in maintaining a clear distinction between **raw and preprocessed data**, which sometimes led to confusion and inconsistent results. Additionally, repeated retraining due to **environment resets (e.g., Colab runtime issues)** introduced variability in results, highlighting the importance of reproducibility practices such as saving model checkpoints and fixing random seeds. Overall, addressing these challenges required a **systematic debugging approach**, including step-by-step validation of each pipeline component—data preprocessing, model training, evaluation, and deployment—ensuring a robust and reliable final system.

Key Learnings

This project provided valuable insights into both the **technical depth** and **practical challenges** of building real-world machine learning systems. One of the most important learnings was the critical role of **consistent data preprocessing** across training and inference stages. Even minor discrepancies in image normalization, resizing, or formatting can significantly degrade model performance, highlighting that a well-trained model is only as reliable as its input pipeline. We also realized that **high accuracy alone is not a sufficient**

indicator of model reliability. A model may achieve strong performance on test data while still being biased or focusing on irrelevant features. This emphasized the importance of **model validation beyond metrics**, particularly in sensitive domains like medical imaging. Techniques such as **Grad-CAM** proved essential for interpreting model decisions, allowing us to visually verify whether the model was focusing on meaningful tumor regions rather than noise or artifacts. Another key takeaway was that **deployment is a distinct and challenging phase** of the machine learning lifecycle. Issues such as input shape mismatches, API integration errors, and environment inconsistencies often arise only after transitioning from development to production. This reinforced the need for robust testing, validation, and standardization before deployment.

Finally, the project highlighted the importance of **iterative development and systematic debugging**. Progress was achieved through continuous experimentation, error analysis, and refinement at each stage of the pipeline. This iterative approach not only improved the model's performance but also deepened our understanding of end-to-end machine learning workflows, making the system more reliable and production-ready.

Conclusion

This project successfully demonstrates the application of deep learning and Explainable AI in medical image analysis. By combining EfficientNet with Grad-CAM, we achieved high accuracy while maintaining interpretability. The comparison with traditional machine learning models highlighted the advantages of deep learning in handling complex image data.

Despite challenges, we developed a complete end-to-end system capable of real-time predictions and visualization. The project emphasizes the importance of consistency, debugging, and iterative improvement. Overall, it provides a strong foundation for future research and development in AI-based medical diagnostics.

Future Work

There are several directions for future improvement. Enhancing Grad-CAM precision to better localize tumors is one potential area. Incorporating additional Explainable AI techniques such as SHAP or LIME could provide deeper insights. Extending the model to handle 3D MRI data would improve clinical relevance.

We also plan to deploy the system on cloud platforms like AWS or GCP for scalability. Building a mobile application could make the system more accessible. Improving dataset diversity and addressing class imbalance could further enhance model performance. Continuous refinement of zreprocessing techniques is another area for exploration.

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